

Approach

The answer is **always computed by a deterministic symbolic solver**, never by a language model. The LLM (**Qwen2.5-3B-Instruct**, Qwen License, $\leq 8B$, self-hosted via Ollama, OpenAI-compatible /v1/*) does three utility roles: (1) extract quantities from prose when regex misses, (2) translate NL claims to FOL for Z3, (3) phrase the explanation. Every numeric result comes from SymPy/pint; every logical verdict from forward-chaining or Z3. The explanation is generated from the solver’s trace, so it cannot diverge from the answer – directly serving P2 (explanation) and P3 (reasoning depth).

Pipeline

- **Router**: presence of **premises-NL** -> logic, else physics.
- **Physics**: cleaner -> topic classifier (ID-prefix + keyword + field-vs-force routing guard + symbol fallback + state-change detector) -> 6-pass role-aware regex extractor (chained equality, noun-of-value, “X cm away from / apart”, “has resistance / is at”, “draws X A”, “doubled/tripled” -> ratio) with LLM fallback (N=3 self-consistency vote + TF-IDF few-shot RAG over solved train rows; LLM strictly forbidden to pre-convert units) -> pint normalisation -> **35-formula YAML registry** (capacitor/Coulomb/parallel-plate variants, E-field from force/point-charge/midpoint, isoceles & right-triangle apex, voltage capacitor distance ratio Q-conserved, magnetic flux/solenoid/wire, RLC impedance/resonance/factor/power, Ohm’s law, passthrough state-change) -> SymPy compute -> sanity verifier -> trace explanation.
- **Logic**: TF-IDF premise selector ($R@5 = 81.3\%$ vs the dataset’s gold `idx` field) -> rule parser -> forward chainer -> **Z3** entailment (multi-variable universals, comparison literals) with a NL->FOL parser/vocabulary rejection-and-retry loop (Slide-27 endorsed) -> per-option entailment for MC with **Unknown** abstention.
- **Self-correction** ON: solver re-checks the LLM draft, feedback is *solver-grounded* (never invents numbers/premises), ≤ 2 rounds. Output schema {**answer**, **explanation**, **cot**, **premises**, **fol**, **confidence**}; malformed JSON impossible (Pydantic); unreachable LLM degrades to the deterministic answer (never a 500).

Training (ablation, not deployed)

QLoRA-SFT on Qwen2.5-7B (`unsloth/qwen2.5-7b-instruct`, rank 16 alpha 32, 2 epochs, Colab T4, 1,765 pairs from 2026-05-09). Measured on a **163-row SFT-unseen** holdout of 2026-05-15 (`scripts/build_sft_unseen_holdout.py` fixes a 99% leakage in our prior holdout): single SFT-7B = 24.5% (-6.6pp LD regression, 5x latency); hybrid LD->3B + rest->SFT-7B = 30.1%. **Not deployed**: Q3 “one model resident” compliance + vLLM-LoRA stack is 1-2 days of risk that does not justify marginal lift over deterministic 33.1%.

Data & compliance

Only the official EXACT2026 release **2026-05-15**. No external/synthetic/crawled data, no GPT/Claude/Gemini in the pipeline. See `docs/data_disclosure.pdf`. 253 automated tests; ruff + mypy CI; 30 ADRs documenting every architectural decision and measurement.

Internal results (163-row SFT-unseen clean holdout, frozen unit-/round-aware scorer)

Variant	Physics Full	Logic correct
Rule-only (Day-1) / Honest baseline (Day-21)	0.8% / 22.1%	35.9% / 22.2%
+ F1+F2 audit formulas + field-vs-force guard / F3+E5 selector+FOL loop	27.6%	23.5%
+ Iter-3 architectural fixes (role-aware regex + state-change + boundary)	33.1%	—
+ hybrid LD->3B + rest->SFT-7B (projected on iter-3, not deployed)	~35.6%	—

Iter-3 lift by prefix (vs 27.6%): DT 0->33%, TD 23->35%, THCB 44->56%, NL 14->24%, LD 33->38%.

Physics correctness uses strict exact value-and-unit match (pint dimension + 1% rel tolerance OR round-aware match at the gold’s stated precision). The deterministic solver is exact where extraction succeeds; the residual is LLM extraction recall on hard multi-step problems. Logic is translator-bound on the 3B; the symbolic Z3 core is sound.

Deployment

`docker compose -f docker/docker-compose.yml up -d` -> GPU Ollama sidecar (qwen2.5:3b pinned resident, `OLLAMA_KEEP_ALIVE=-1`) + CPU FastAPI solver; always-on with `/healthz`, cron pre-warm (`scripts/prewarm_cron.sh`). Reproduce: `uv run pytest -q; uv run python scripts/run_eval.py --task {physics,logic} --with-llm --split data/official_v20260515/eval_split/<task>_eval.jsonl`.